Prisha

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Education

Vellore Institute of Technology, Bhopal — Sep 2023 – Present
B.Tech in Computer Science (AI & ML) — *CGPA: 9.13 / 10*

Anand Public Sr Sec School, Kapurthala — 2023
Class XII — *89%*

Christ King Convent School, Kapurthala — 2021
Class X — *94%*

Technical Skills

Languages: Python, C++, Java
Python Ecosystem: NumPy, Pandas, NetworkX, PyVis, Scikit-learn, PyTorch Geometric, Flask, Tkinter, Subprocess
Core CS: Data Structures and Algorithms, OOP, Graph Theory, Clustering Algorithms
Systems and Tools: Linux, WSL2, Git, MySQL, REST APIs, AWS (EC2, S3, IAM)
Web: HTML, CSS, Jinja2

Experience

FOSSEE, IIT Bombay — Feb 2026 – May 2026
Python Developer, Semester Long Intern, eSim KiCad Plugin (Remote/Hybrid)

- Built **PCB Linter**, a 1500+ line Python tool automating PCB circuit analysis using graph algorithms and clustering; processes KiCad board files via Python scripting API
- Implemented **Deterministic Annealing clustering** from scratch in Python; auto-groups circuit nodes into voltage, current, and power domains using similarity matrices and temperature-based annealing
- Engineered a **SPICE netlist generator** in Python that parses PCB component data, sanitises net names, maps 10+ component types, and auto-injects subcircuit models from a 647-file library
- Built a **universal file parser** in Python handling 4 simulation output formats (DC, AC, Transient, CSV) with automatic format detection using regex and string processing
- Used **NetworkX** for PCB connectivity graph construction; computed degree centrality and betweenness centrality to detect high-fanout nodes and single points of failure
- Generated dynamic **HTML reports** via Python string templating with interactive clustering tabs, heatmaps, voltage attenuation matrices, and dark/light mode toggle
- Automated end-to-end pipeline: PCB parsing, SPICE generation, NGSpice subprocess execution, output parsing, clustering, and report generation, entirely in Python

FOSSEE, IIT Bombay — May 2026 – Present
Python Developer, eSim Summer Fellow, KiCad Plugin (Remote/Hybrid)

Projects

eSim Tool Manager | *Python, subprocess, apt, winget* — Jan 2026 – Feb 2026

- Built a cross-platform Python CLI tool for automated installation and configuration of eSim and all dependencies on Linux and Windows
- Integrated apt and winget package managers via Python subprocess calls; implemented version detection, PATH configuration, and automated upgrade workflows
- Reduced manual setup time significantly; adopted by FOSSEE team for intern onboarding

Power Consumption Analysis | *Python, Flask, Scikit-learn, Matplotlib* — May 2025 – Jul 2025

- Built an ML-powered web dashboard using Flask to analyze and visualize household energy consumption patterns
- Performed data preprocessing, feature engineering, and trained regression models to identify consumption inefficiencies
- Deployed interactive charts using Matplotlib and Jinja2 templates for real-time data exploration

Crop Recommendation System | *Python, Scikit-learn, Pandas* — Feb 2025 – Apr 2025

- Developed ML model for crop recommendation based on soil and climate data; improved accuracy using feature engineering and cross-validation
- Built data pipeline in Python for cleaning, encoding, and transforming agricultural datasets for model training

Achievements & Certifications

Machine Learning Specialization, University of Michigan (Coursera) | Microsoft Azure Data Fundamentals | IBM SkillsBuild AI Certification | AWS Cloud Practitioner and Solutions Architect | Solved 100+ DSA problems on LeetCode

Leadership

Cricket Team Captain, VIT Bhopal | Participant, All India ICSE Frank Anthony Debate Competition | Head Girl, Christ King Convent School